Leak test (ntuple clusters, file 3, first 20 events): AUC(unlinked vs linked) gain of the added columns over the F1 probe (shape + position) and the strong probe (F1 + tiny-CNN mean/sigma + CPE offset); HGB trained on even events, tested on odd (and swapped). No-angle convention for the payload: an absent angle = NaN

test	group	n	frac_unlinked	F1_base	F1_gain	F1_gain_swapped	strong_base	strong_gain	strong_gain_swapped
emu: emitted angles only (unchanged by the table)	IL2-4	363950	0.1287	0.7762	0.009514	0.01107	0.8012	0.0002187	-1.928e-05
emu: emitted angles only (unchanged by the table)	IL1	180648	0.1057	0.8017	0.0009529	0.001411	0.8204	0.001581	-0.0008817
emu: emitted angles only (unchanged by the table)	ID1-4	910220	0.1829	0.7518	-0.0005538	-0.0004332	0.7541	3.428e-05	-0.0001272
emu: emitted angles + sigmas (v2 table)	IL2-4	363950	0.1287	0.7762	0.008971	0.01038	0.8012	-0.0006921	-0.001573
emu: emitted angles + sigmas (v2 table)	IL1	180648	0.1057	0.8017	0.00306	0.0003228	0.8204	0.0005672	9.741e-06
emu: emitted angles + sigmas (v2 table)	ID1-4	910220	0.1829	0.7518	-0.0003534	-0.0005319	0.7541	0.0001968	-0.0002456
emu: emitted angles + sigmas (v3 table)	IL2-4	363950	0.1287	0.7762	0.00925	0.01017	0.8012	-0.0002263	-0.0003524
emu: emitted angles + sigmas (v3 table)	IL1	180648	0.1057	0.8017	0.002196	-0.0004096	0.8204	0.001078	-0.0006925
emu: emitted angles + sigmas (v3 table)	ID1-4	910220	0.1829	0.7518	-0.0002444	-0.0002176	0.7541	-0.0002715	-0.0001805
payload v4 (old units), old code: angles + sigmas + has*	IL2-4	363950	0.1287	0.7762	0.1531	0.1509	0.8012	0.1399	0.1372
payload v4 (old units), old code: angles + sigmas + has*	IL1	180648	0.1057	0.8017	0.1514	0.1482	0.8204	0.1398	0.1419
payload v4 (old units), old code: angles + sigmas + has*	ID1-4	910220	0.1829	0.7518	0	0	0.7541	0	0
payload v5 (physical units), new code: angles + sigmas + has*	IL2-4	363950	0.1287	0.7762	0.1479	0.1504	0.8012	0.1401	0.136
payload v5 (physical units), new code: angles + sigmas + has*	IL1	180648	0.1057	0.8017	0.1482	0.1451	0.8204	0.1354	0.1403
payload v5 (physical units), new code: angles + sigmas + has*	ID1-4	910220	0.1829	0.7518	0	0	0.7541	0	0

Leak test (nano, file 3, 100 events): gain of ALL angle columns (cot, sigma, has, sat) over the nano probe

test	group	n	auc_no_angles	auc_with_all_angle_columns	gain
emu nano, v2 table (all angle columns incl. sigma)	IL2-4	1815061	0.7818	0.791	0.009192
emu nano, v2 table (all angle columns incl. sigma)	IL1	899091	0.8166	0.8207	0.004155
emu nano, v2 table (all angle columns incl. sigma)	ID1-4	4532754	0.7529	0.7525	-0.0003757
emu nano, v3 table (all angle columns incl. sigma)	IL2-4	1815061	0.7818	0.7906	0.008732
emu nano, v3 table (all angle columns incl. sigma)	IL1	899091	0.8166	0.8221	0.005488
emu nano, v3 table (all angle columns incl. sigma)	ID1-4	4532754	0.7529	0.7524	-0.0004754
payload v4 (old units), old code nano (all angle columns)	IL2-4	1815061	0.7818	0.9451	0.1632
payload v4 (old units), old code nano (all angle columns)	IL1	899091	0.8166	0.9731	0.1566
payload v4 (old units), old code nano (all angle columns)	ID1-4	4532754	0.7529	0.7529	0
payload v5 (physical units), new code nano (all angle columns)	IL2-4	1815061	0.7818	0.9468	0.165
payload v5 (physical units), new code nano (all angle columns)	IL1	899091	0.8166	0.9722	0.1557
payload v5 (physical units), new code nano (all angle columns)	ID1-4	4532754	0.7529	0.7529	0